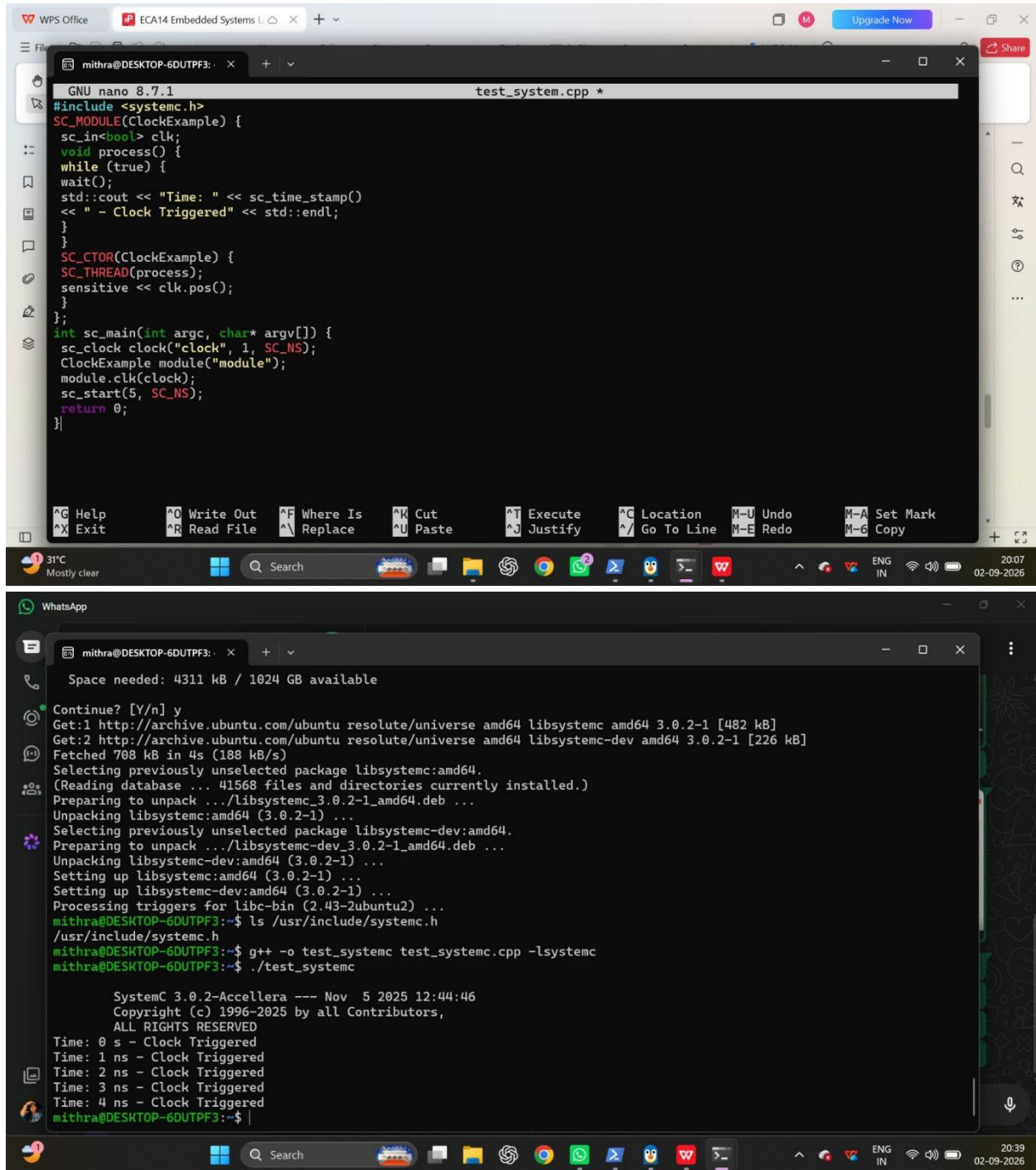


Exp No: 24 MODELING CLOCK-DRIVEN CONCURRENT PROCESSES USING SC_THREAD AND WAIT() IN SYSTEMC



```
GNU nano 8.7.1 test_system.cpp *
#include <systemc.h>
SC_MODULE(ClockExample) {
    sc_in<bool> clk;
    void process() {
        while (true) {
            wait();
            std::cout << "Time: " << sc_time_stamp()
            << " - Clock Triggered" << std::endl;
        }
    }
    SC_CTOR(ClockExample) {
        SC_THREAD(process);
        sensitive << clk.pos();
    }
};

int sc_main(int argc, char* argv[]) {
    sc_clock clock("clock", 1, SC_NS);
    ClockExample module("module");
    module.clk(clock);
    sc_start(5, SC_NS);
    return 0;
}

Space needed: 4311 kB / 1024 GB available
Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libsystemc amd64 3.0.2-1 [482 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/universe amd64 libsystemc-dev amd64 3.0.2-1 [226 kB]
Fetched 708 kB in 4s (188 kB/s)
Selecting previously unselected package libsystemc:amd64.
(Reading database ... 41568 files and directories currently installed.)
Preparing to unpack .../libsystemc_3.0.2-1_amd64.deb ...
Unpacking libsystemc:amd64 (3.0.2-1) ...
Selecting previously unselected package libsystemc-dev:amd64.
Preparing to unpack .../libsystemc-dev_3.0.2-1_amd64.deb ...
Unpacking libsystemc-dev:amd64 (3.0.2-1) ...
Setting up libsystemc:amd64 (3.0.2-1) ...
Setting up libsystemc-dev:amd64 (3.0.2-1) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
mithra@DESKTOP-6DUTPF3:~$ ls /usr/include/systemc.h
/usr/include/systemc.h
mithra@DESKTOP-6DUTPF3:~$ g++ -o test_systemc test_systemc.cpp -lsystemc
mithra@DESKTOP-6DUTPF3:~$ ./test_systemc

SystemC 3.0.2-Accellera --- Nov 5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
mithra@DESKTOP-6DUTPF3:~$
```